

A Staircase Perturbation Proves the Uniqueness Conjecture for Independent Copies in L_p

Candidate proof packet for human verification

Run `fa_banach_001`, lane 2, model GPT5.6

9 August 2026

Status: candidate full solution, likely valid. This packet claims an affirmative proof of Conjecture 2 of Astashkin–Sukochev–Zanin. It has not been human verified, and no public novelty claim should be made before expert review and an independent citation search.

Abstract

Let $1 \leq p < 2$ and let M be a p -convex and 2-concave Orlicz function. Astashkin, Sukochev, and Zanin conjectured that uniqueness, up to equivalence near zero, of the distribution of a mean-zero random variable whose independent copies are equivalent in L_p to the unit vector basis of ℓ_M forces M to be $(p + \varepsilon)$ -convex and $(2 - \varepsilon)$ -concave for some $\varepsilon > 0$. We prove the conjecture. The source paper reduces the generator condition to comparability of a two-term Hardy envelope with $\mathbf{m}(t) = 1/M^{-1}(t)$. Unless a given generator is already comparable with $\mathbf{m}$, we add a thinned decreasing staircase whose Hardy envelope remains dominated by $\mathbf{m}$ but whose ratio to the original generator is unbounded on positive-measure intervals. This produces a second non-equivalent generating distribution and contradicts uniqueness.

1 Source problem

The source is S. Astashkin, F. Sukochev, and D. Zanin, *On uniqueness of distribution of a random variable whose independent copies span a subspace in L_p* , *Studia Mathematica* 230 (2015), 41–57, arXiv:1406.4950 [1]. The conjecture appears on page 3 of the source PDF.

Transcription of Conjecture 2. Let $1 \leq p < 2$ and let M be a p -convex and 2-concave Orlicz function. If there is a unique (up to equivalence near 0) mean zero function f whose independent copies are equivalent in L_p to the unit vector basis in ℓ_M , then M is $(p + \varepsilon)$ -convex and $(2 - \varepsilon)$ -concave for some $\varepsilon > 0$.

2 Definitions and source reductions

Throughout, $M(1) = 1$ and M is a bijection from $[0, \infty)$ onto itself, as in the source. An Orlicz function M is r -convex when $s \mapsto M(s^{1/r})$ is convex and is r -concave when this map is concave. For nonnegative functions x, y , the notation $x \asymp y$ means two-sided comparability by a positive constant. Equivalence near zero means that this comparison holds for all sufficiently small positive arguments.

The proof of Theorem 1 is presented in Section 4. Two important components of the proof are Proposition 6 and Theorem 9 which are given in Sections 2 and 3 respectively.

We propose the following conjecture.

Conjecture 2. *Let $1 \leq p < 2$ and let M be a p -convex and 2-concave Orlicz function. If there is a unique (up to equivalence near 0) mean zero function f whose independent copies are equivalent in L_p to the unit vector basis in l_M , then M is $(p + \varepsilon)$ -convex and $(2 - \varepsilon)$ -concave for some $\varepsilon > 0$.*

2. PRELIMINARIES AND AUXILIARY RESULTS

2.1. Orlicz functions and spaces. For the theory of Orlicz spaces we refer to [17].

Figure 1: The complete statement of Conjecture 2 in the source paper, rendered from source PDF page 3.

For a nonnegative decreasing x on $(0, 1)$, set

$$T_p x(t) := \left(\frac{1}{t} \int_0^t x(s)^p ds \right)^{1/p} + \left(\frac{1}{t} \int_t^1 x(s)^2 ds \right)^{1/2}, \quad 0 < t \leq 1, \quad (1)$$

and put

$$\mathbf{m}(t) := \frac{1}{M^{-1}(t)}. \quad (2)$$

We use two results proved in the source paper. Its Proposition 6 says that if $f \in L_p(0, 1)$ is mean zero, then its independent copies are equivalent in L_p to the unit vector basis of ℓ_M if and only if

$$T_p(f^*) \asymp \mathbf{m} \quad (0 < t \leq 1), \quad (3)$$

where f^* is the decreasing rearrangement of $|f|$. Its Theorem 9 says, under the present p -convexity and 2-concavity assumptions, that

$$T_p \mathbf{m} \asymp \mathbf{m} \iff M \text{ is } (p + \varepsilon)\text{-convex and } (2 - \varepsilon)\text{-concave for some } \varepsilon > 0. \quad (4)$$

Two elementary monotonicities will drive the construction. Define

$$P(t) := t\mathbf{m}(t)^p, \quad Q(t) := t\mathbf{m}(t)^2. \quad (5)$$

Writing $t = M(s)$ gives

$$P(M(s)) = \frac{M(s)}{s^p}, \quad Q(M(s)) = \frac{M(s)}{s^2}.$$

Convexity of $u \mapsto M(u^{1/p})$ and its vanishing at zero imply that $M(s)/s^p$ is increasing. Concavity of $u \mapsto M(u^{1/2})$ implies that $M(s)/s^2$ is decreasing. Thus

$$P \text{ is increasing,} \quad Q \text{ is decreasing.} \quad (6)$$

3 Idea of the proof

Suppose $h = f^*$ is one generating rearrangement. By (3), $T_p h \asymp \mathfrak{m}$, and monotonicity of h immediately gives $h \lesssim \mathfrak{m}$. If the reverse comparison holds near zero, then each Hardy term of $T_p \mathfrak{m}$ is controlled by the corresponding term for h ; Theorem 9 then supplies the desired strict indices.

The only remaining case is that $\mathfrak{m}/h$ is unbounded arbitrarily close to zero. At carefully thinned continuity points $t_n \downarrow 0$, put a staircase level

$$a_n = 2^{-n} \mathfrak{m}(t_n)$$

on $[t_{n+1}, t_n]$. The thinning makes (a_n) increase with n , so the staircase is decreasing as a function of t . The geometric coefficient 2^{-n} makes the p -head summable against the increasing quantity P and the 2-tail summable against the decreasing quantity Q . Consequently the staircase u satisfies $T_p u \lesssim \mathfrak{m}$. Yet the points t_n are chosen where h is much smaller than a_n , and continuity makes this domination persist on intervals of positive length. Hence $g = h + u$ has the same admissible Hardy envelope as h but is not equivalent to h near zero. An explicit sign symmetrization realizes g as the rearrangement of a mean-zero random variable, giving the forbidden second generator.

4 The staircase lemma

Lemma 1 (Admissible non-equivalent perturbation). *Let $1 \leq p < 2$, let M be p -convex and 2-concave, and let $\mathfrak{m}$ be given by (2). Suppose $h \in L_p(0, 1)$ is nonnegative, decreasing, and*

$$h \lesssim \mathfrak{m}. \quad (7)$$

If $\mathfrak{m}/h$ is unbounded in every neighborhood of zero, then there is a nonnegative decreasing $u \in L_p(0, 1)$ such that

$$T_p u \lesssim \mathfrak{m}, \quad (8)$$

and $(h + u)/h$ is unbounded on positive-measure intervals accumulating at zero.

Proof. Because $\mathfrak{m}(t) \rightarrow \infty$ as $t \downarrow 0$, and because $\mathfrak{m}/h$ is unbounded in every such neighborhood, we may inductively choose continuity points

$$t_1 > t_2 > \cdots \downarrow 0$$

of the monotone function h so that, for $n \geq 2$,

$$t_n < \frac{1}{2} t_{n-1}, \quad \mathfrak{m}(t_n) \geq 2\mathfrak{m}(t_{n-1}), \quad h(t_n) \leq \frac{2^{-n-2}}{n} \mathfrak{m}(t_n). \quad (9)$$

The same last inequality is imposed for $n = 1$. Here one may first find a point with a sufficiently large ratio and then move slightly to its right to a continuity point; right continuity of h and continuity of $\mathfrak{m}$ preserve the strict inequality. After choosing t_n , continuity lets us fix $0 < \delta_n < t_n/2$ such that

$$h(s) \leq \frac{2^{-n-1}}{n} \mathfrak{m}(t_n) \quad \text{for } t_n - \delta_n < s < t_n. \quad (10)$$

At the next induction step we additionally require $t_{n+1} < t_n - \delta_n$.

Set

$$a_n := 2^{-n} \mathfrak{m}(t_n). \quad (11)$$

The middle condition in (9) gives $a_{n+1} \geq a_n$. Define the right-continuous decreasing staircase, up to irrelevant endpoint values, by

$$u(t) := \begin{cases} a_n, & t_{n+1} \leq t < t_n, \quad n \geq 1, \\ 0, & t_1 \leq t \leq 1. \end{cases} \quad (12)$$

The intervals cover $(0, t_1)$ because $t_n \downarrow 0$.

First, $u \in L_p$. Indeed, using that P is increasing,

$$\begin{aligned} \int_0^1 u(t)^p dt &\leq \sum_{n \geq 1} t_n a_n^p = \sum_{n \geq 1} 2^{-np} P(t_n) \\ &\leq P(t_1) \sum_{n \geq 1} 2^{-np} < \infty. \end{aligned} \quad (13)$$

Fix $t \in [t_{n+1}, t_n)$. For the head term, again by monotonicity of P ,

$$\begin{aligned} \int_0^t u(s)^p ds &\leq \sum_{k \geq n+1} t_k a_k^p + t a_n^p \\ &= \sum_{k \geq n+1} 2^{-kp} P(t_k) + t 2^{-np} \mathbf{m}(t_n)^p \\ &\leq P(t) \sum_{k \geq n+1} 2^{-kp} + t \mathbf{m}(t)^p \lesssim t \mathbf{m}(t)^p. \end{aligned} \quad (14)$$

Here $t_k \leq t$ for $k \geq n+1$, while $t < t_n$ and the decrease of $\mathbf{m}$ give $\mathbf{m}(t_n) \leq \mathbf{m}(t)$.

For the tail term, the decreasing character of Q yields

$$\begin{aligned} \int_t^1 u(s)^2 ds &\leq \sum_{k < n} t_k a_k^2 + t_n a_n^2 \\ &= \sum_{k < n} 2^{-2k} Q(t_k) + 2^{-2n} Q(t_n) \\ &\leq Q(t) \left(\sum_{k < n} 2^{-2k} + 2^{-2n} \right) \lesssim t \mathbf{m}(t)^2. \end{aligned} \quad (15)$$

Indeed $t_k > t$ for $k < n$ and $t_n > t$, so $Q(t_k), Q(t_n) \leq Q(t)$. If $t \geq t_1$, the tail vanishes and (13), together with $P(t_1) \leq P(t)$, gives the same head estimate. Equations (14) and (15) prove (8).

Finally, the interval $(t_n - \delta_n, t_n)$ lies inside $[t_{n+1}, t_n)$, so (10) gives there

$$\frac{u(s)}{h(s)} \geq \frac{2^{-n} \mathbf{m}(t_n)}{2^{-n-1} \mathbf{m}(t_n)/n} = 2n.$$

These are positive-length intervals and they accumulate at zero. Hence $(h+u)/h$ is not bounded near zero. $\square$

5 Full solution of the conjecture

Theorem 2 (Affirmative answer to Conjecture 2). *Let $1 \leq p < 2$ and let M be a p -convex and 2-concave Orlicz function. If there is a unique, up to equivalence of decreasing rearrangements near zero, mean-zero function whose independent copies are equivalent in L_p to the unit vector basis of ℓ_M , then M is $(p + \varepsilon)$ -convex and $(2 - \varepsilon)$ -concave for some $\varepsilon > 0$.*

Proof. Let $f \in L_p(0, 1)$ be the generator supplied by the hypothesis and set $h = f^*$. By Proposition 6 of the source paper,

$$T_p h \asymp \mathbf{m}. \quad (16)$$

Since h is decreasing,

$$h(t) \leq \left(\frac{1}{t} \int_0^t h(s)^p ds \right)^{1/p} \lesssim \mathbf{m}(t),$$

so (7) holds.

We first dispose of the case in which $\mathbf{m} \lesssim h$ near zero. Choose $t_0 > 0$ so that this comparison holds on $(0, t_0]$. For $t \leq t_0$, its p -head gives

$$\int_0^t \mathbf{m}(s)^p ds \lesssim \int_0^t h(s)^p ds \lesssim t \mathbf{m}(t)^p.$$

For the 2-tail, split at t_0 :

$$\begin{aligned} \int_t^1 \mathbf{m}(s)^2 ds &\lesssim \int_t^{t_0} h(s)^2 ds + \int_{t_0}^1 \mathbf{m}(s)^2 ds \\ &\lesssim t \mathbf{m}(t)^2 + C_0. \end{aligned}$$

The constant C_0 is also bounded by a multiple of $t \mathbf{m}(t)^2 = Q(t)$, because Q is decreasing and $t \leq t_0$. Hence $T_p \mathbf{m} \lesssim \mathbf{m}$ for small t . On $[t_0, 1]$ the same bound follows from compactness, while the reverse inequality

$$\mathbf{m}(t) \leq \left(\frac{1}{t} \int_0^t \mathbf{m}(s)^p ds \right)^{1/p}$$

is automatic since $\mathbf{m}$ is decreasing. Thus $T_p \mathbf{m} \asymp \mathbf{m}$. The source paper's Theorem 9, (4), proves the desired strict indices.

Suppose, toward a contradiction, that the strict indices fail. The preceding paragraph and $h \lesssim \mathbf{m}$ show that $\mathbf{m}/h$ must be unbounded in every neighborhood of zero. Apply Lemma 1, and set

$$g := h + u.$$

Then g is nonnegative, decreasing, and belongs to $L_p(0, 1)$. Minkowski's inequality in each of the two terms defining T_p gives

$$T_p g \leq T_p h + T_p u \lesssim \mathbf{m},$$

whereas positivity gives $T_p g \geq T_p h \gtrsim \mathbf{m}$. Therefore

$$T_p g \asymp \mathbf{m}. \quad (17)$$

It remains only to realize g as the decreasing rearrangement of a mean-zero random variable on $(0, 1)$. Define, ignoring the single midpoint,

$$y(t) := \begin{cases} g(2t), & 0 < t < \frac{1}{2}, \\ -g(2t - 1), & \frac{1}{2} < t < 1. \end{cases} \quad (18)$$

Then $y \in L_p(0, 1)$, the two halves cancel so that $\int_0^1 y = 0$, and $|y|$ has exactly the same distribution as g . Thus $y^* = g$. By (17) and source Proposition 6, the independent copies of y are also equivalent in L_p to the unit vector basis of ℓ_M .

Lemma 1 says that g/h is unbounded on positive-measure intervals accumulating at zero. Consequently g and h are not equivalent near zero. The mean-zero random variables y and f therefore give two non-equivalent generating distributions, contrary to the uniqueness hypothesis. The strict indices cannot fail, and the conjecture follows. $\square$

Remark 3 (Constructive strengthening). The proof gives slightly more than the contrapositive. Whenever a generating rearrangement h is not comparable with $\mathbf{m}$ near zero, it explicitly constructs a second generating rearrangement $g \geq h$ for which g/h is unbounded on intervals accumulating at zero.

6 Verification and computational check

The argument is purely analytic. The included script `code/check_staircase_bounds.py` is only a regression check for the head and tail bookkeeping. It uses power envelopes $\mathbf{m}(t) = t^{-1/q}$ for four pairs $1 \leq p < q < 2$, three geometric meshes, and 4000 logarithmic sample points per case. It verifies monotonic staircase levels, finite L_p mass, and a finite sampled upper bound for $T_p u / \mathbf{m}$. Run it from the repository root with

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
  1406.4950_uniqueness_conjecture_staircase_perturbation/\
  code/check_staircase_bounds.py
```

The symbolic estimates (14)–(15), not the finite test, prove the lemma. A detailed adversarial audit appears in `verification.md`.

7 Scope, novelty check, and limitations

This packet answers the exact pointwise near-zero equivalence formulation in Conjecture 2. It does not claim a classification for stronger notions of quasi-equivalence that allow rescaling the argument of distribution functions.

A bounded literature screen was performed on 9 August 2026 before promotion. The repository’s lightweight indexes were searched for the arXiv id, title, and core terms. External searches used the exact title, “Conjecture 2,” the three authors, and combinations of “uniqueness distribution,” “independent copies,” “Orlicz,” “ p -convex,” and “2-concave.” No later paper claiming this converse and no matching staircase proof was found.

The strongest later-status checks were the same authors’ 2022 follow-up [2], which describes the global uniqueness question as remaining open while proving further sufficient theorems, and Astashkin’s 2024 survey [3], whose Problem 1 again asks for necessary and sufficient uniqueness conditions. The survey’s formulation uses quasi-equivalence, so it is supporting evidence that the surrounding problem remained active, not an exact identity with Conjecture 2. This was a bounded search, not a proof of novelty.

The recommended human review should focus on: (i) the two monotonicity directions in (6); (ii) the arbitrary- t estimates (14) and (15); (iii) selection of continuity points and positive-length intervals in Lemma 1; and (iv) the exact fit between the symmetrization (18) and source Proposition 6.

References

- [1] S. Astashkin, F. Sukochev, and D. Zanin, *On uniqueness of distribution of a random variable whose independent copies span a subspace in L_p* , *Studia Math.* **230** (2015), no. 1, 41–57. arXiv:1406.4950; doi:10.4064/sm8089-1-2016.

- [2] S. Astashkin, F. Sukochev, and D. Zanin, *The distribution of a random variable whose independent copies span ℓ_M is unique*, Rev. Mat. Complut. **35** (2022), 815–834. doi:10.1007/s13163-021-00406-x.
- [3] S. V. Astashkin, *Sequences of independent functions and structure of rearrangement invariant spaces*, Russian Math. Surveys **79** (2024), no. 3, 375–457. Math-Net.Ru record.